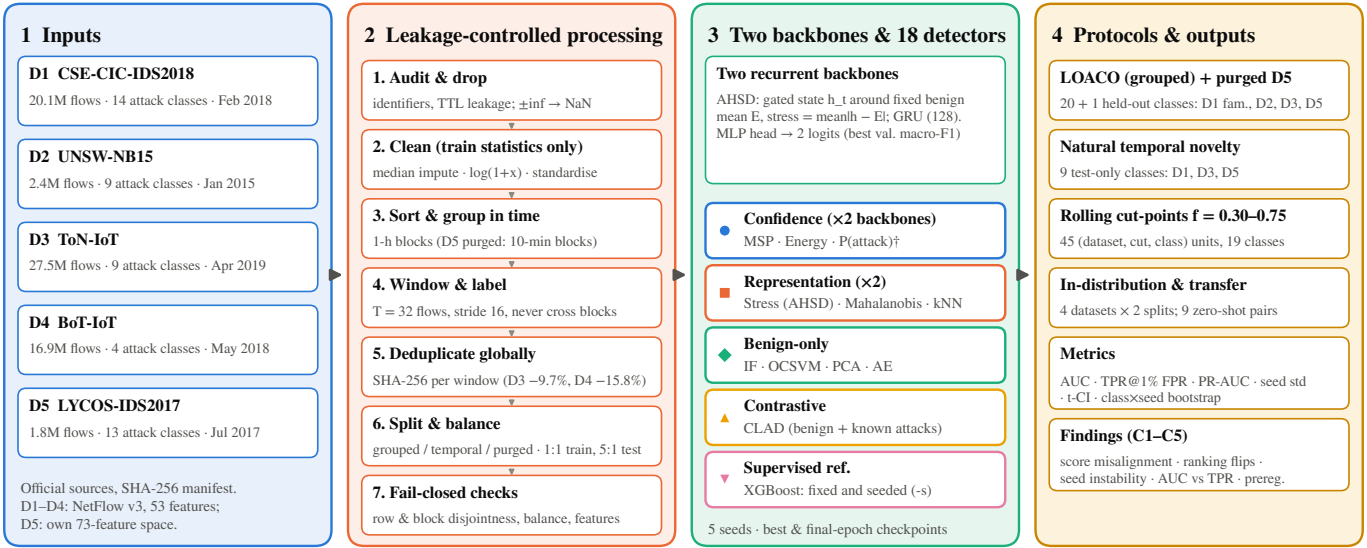
1 Inputs
D1 CSE-CIC-IDS2018
20.1M flows · 14 attack classes · Feb 2018
D2 UNSW-NB15
2.4M flows · 9 attack classes · Jan 2015
D3 ToN-IoT
27.5M flows · 9 attack classes · Apr 2019
D4 BoT-IoT
16.9M flows · 4 attack classes · May 2018
D5 LYCOS-IDS2017
1.8M flows · 13 attack classes · Jul 2017
Official sources, SHA-256 manifest.
D1–D4: NetFlow v3, 53 features;
D5: own 73-feature space.
2 Leakage-controlled processing
1. Audit & drop
identifiers, TTL leakage; ±inf → NaN
2. Clean (train statistics only)
median impute · log(1+x) · standardise
3. Sort & group in time
1-h blocks (D5 purged: 10-min blocks)
4. Window & label
T = 32 flows, stride 16, never cross blocks
5. Deduplicate globally
SHA-256 per window (D3 −9.7%, D4 −15.8%)
6. Split & balance
grouped / temporal / purged · 1:1 train, 5:1 test
7. Fail-closed checks
row & block disjointness, balance, features
3 Two backbones & 18 detectors
Two recurrent backbones
AHSD: gated state h_t around fixed benign
mean E, stress = meanlh − El; GRU (128).
MLP head → 2 logits (best val. macro-F1)
Confidence (×2 backbones)
MSP · Energy · P(attack)†
Representation (×2)
Stress (AHSD) · Mahalanobis · kNN
Benign-only
IF · OCSVM · PCA · AE
Contrastive
CLAD (benign + known attacks)
Supervised ref.
XGBoost: fixed and seeded (-s)
5 seeds · best & final-epoch checkpoints
4 Protocols & outputs
LOACO (grouped) + purged D5
20 + 1 held-out classes: D1 fam., D2, D3, D5
Natural temporal novelty
9 test-only classes: D1, D3, D5
Rolling cut-points f = 0.30–0.75
45 (dataset, cut, class) units, 19 classes
In-distribution & transfer
4 datasets × 2 splits; 9 zero-shot pairs
Metrics
AUC · TPR@1% FPR · PR-AUC · seed std
· t-CI · class×seed bootstrap
Findings (C1–C5)
score misalignment · ranking flips ·
seed instability · AUC vs TPR · prereg.